

1.1 The Structure of Data

■ Outline

- ➊ What is Statistics?
- ➋ Two Areas of Statistics
- ➌ Population versus Sample
- ➍ Variables
 - Qualitative (Categorical) Variables
 - Quantitative (Numerical) Variables
- ➎ Explanatory and response variables

Some Thoughts on Statistics

- Statistics courses are viewed by most college students as an obstacle standing in the way of attaining their desired degree.
- Most consider the statistics course as a painful rite of passage. At the end of the semester, most of those who study it happily put probability-statistics out of their minds forever.
- Nickname for the statistics course: “*Sadistics*”

Formal Definitions of Statistics

- **Statistics** is:

- ① A set of scientific principles and techniques that are useful in reaching conclusions about populations and processes when available information is limited and variable.
- ② A body of principles for designing the process of data collection and making inferences about the population from the information in the sample.
- ③ The analysis of numerical data for the purpose of reaching a decision or communicating information in the face of uncertainty.

- However, these definitions might be too general to be familiar with the actual statistics.
- Considering *how statistics is being used in our real life* would give us a more concrete answer to “what is statistics?”

Practical Understanding of Statistics

- The appearance of statistics tends to be different depending on where you are.
- For example,
 - 1 If you are working for a credit company, one of the most important issues is how the company will decide the consumers' credit level.
 - 2 Pollsters like to know who is going to be the next president in their country.
- People in different areas use statistics in different ways.
- “What is Statistics?” – This question can be better answered, once all your answers to the question are gathered.

Broader Areas of Statistics

- Two areas of statistics:

- 1 **Descriptive statistics** includes *collection*, *reprocessing*, *visualization*, and *summarization*, *description* of *sampled data*.
- 2 **Inferential statistics** refers to the technique of *making decisions* and *drawing conclusions* about *populations*.

Terminology 1.1: Populations

- A **unit (case)** is a *single entity*, usually a person or an object, whose characteristics are of interest.
- The **population** is the *complete collection of units (cases)* about which information is sought.
 - 1 A **finite** population contains a *finite* number of units (cases).
 - 2 A **infinite** population contains an *infinite* number of units (cases).
- Examples:
 - 1 All daily newspapers published in the United States
 - 2 All companies listed on the New York Stock Exchange
 - 3 All students admitted to NAU for Fall 2026
 - 4 All books to be published after 2026

Terminology 1.2: Samples

- A **sample** is a *subset* of a population.
 - 1 A sample plays a role as a *representative* of the population.
 - 2 A **sample of size n** contains n units.
- Example: Give one possible sample of size $n = 5$ from each of the following populations:
 - 1 All daily newspapers published in the United States
 - 2 All companies listed on the New York Stock Exchange
 - 3 All students admitted to NAU for Fall 2026
 - 4 All books to be published after 2026

Terminology 2: Variables

- A **variable** is a *characteristic* about each unit (case) of a population or sample.
- Note: Typically, there are several characteristics or variables of interest for each unit.

Population	Unit	Variables/Characteristics
Registered voters in Arizona	voter	ZIP code age sex conservative/liberal voted or not in last election
All rental apartments in Flagstaff	apartment	monthly rent size in square feet number of bedrooms number of bathrooms Internet connection
All fast food restaurants in Flagstaff	restaurant	number of employees seating capacity

Data Example 1.1

■ Countries of the World:

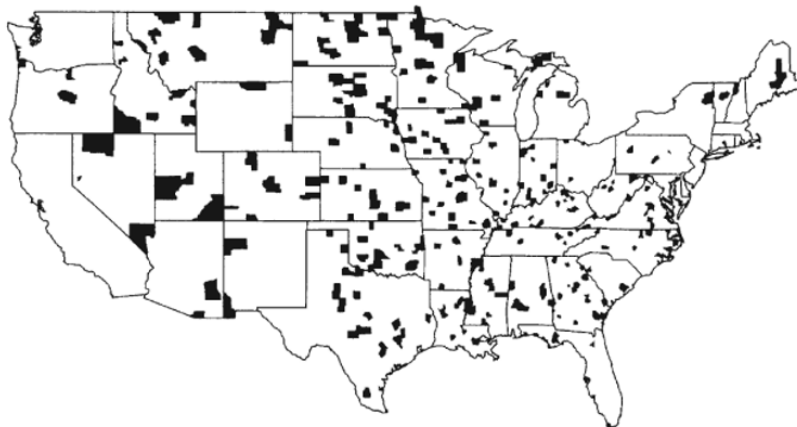
Country	LandArea	Population	Density	GDP	Rural	CO2	PumpPrice	Military
Afghanistan	652.86	30.552	46.8	665	74.1	35.3	1.28	8.65
Albania	27.40	2.897	105.7	4460	44.6	12.8	1.81	
Algeria	2381.74	39.208	16.5	5361	30.5	24.6	0.29	
American Samoa	0.20	0.055	275.0		12.7			
Andorra	0.47	0.079	168.1		13.8	9.5	1.67	
Angola	1246.70	21.472	17.2	5783	57.5	44.8	0.63	13.81
Antigua and Barbuda	0.4	0.090	204.5	13342	75.4	16.6		
Argentina	2736.690	41.446	15.1	14715	8.5	16.9	1.46	

Source: Lock⁵ (2021), John Wiley & Sons, Inc.

■ Note: Generally, each case makes up a row in a dataset, and each variable makes up a column.

Data Example 1.2

- US counties with the highest kidney cancer death rates:



Source: Gelman et al. (2024), *Bayesian Data Analysis*, CRC Press

Terminology 2.1: Qualitative (Categorical) Variables

- A **qualitative (categorical) variable** yields **non-numerical** information.
- Notes:
 - ➊ Arithmetic operations, such as addition and averaging, are **not** meaningful for data resulting from a qualitative (categorical) variable.
 - ➋ Often, qualitative (categorical) variables are coded as numerical values for convenience.
 - Examples: tested positive = 1, negative = 0; male = 0, female = 1; ZIP code.
- Two types of qualitative (categorical) variables:
 - ➊ A **nominal** variable places a unit into one of several *groups* or *categories*.
 - Examples: gender, breeds of animals, colors, brand names of products, so on.
 - ➋ An **ordinal** variable incorporates an *ordered position* or *ranking*.
 - Examples: level of preference, level of satisfaction, ranking, so on.

Terminology 2.2: Quantitative (Numerical) Variables

- A **quantitative (numerical) variable** yields *numerical* measurements.
- Note: *Arithmetic operations are meaningful* for data resulting from a quantitative variable.
- Two types of quantitative (numerical) variables:
 - 1 A **discrete** variable can assume only a *countable* number of values.
 - *Counts* are discrete variables.
 - Examples: number of defects, number of students registered in courses.
 - 2 A **continuous** variable can take any one of an *uncountable* number of values in an interval.
 - *Measurements* are continuous variables.
 - Examples: height, weight, birth rate, SAT score, GPA score, temperature, wind speed.
- *Note*: Distinguishing between categorical and quantitative variables (more specifically, nominal, ordinal, discrete, and continuous) is important in statistics, as the appropriate analysis methods depend on the type of variable.

Exercise 1.1

- Identify each of the following as an example of categorical (nominal, ordinal) or quantitative (discrete, continuous) variables.
 - 1 the taste order of a chocolate pie
 - 2 the breaking strength of a given type of string
 - 3 the species of an insect found in a plot of cropland
 - 4 the temperature in Flagstaff at noon on a given Sunday
 - 5 whether or not a 60 watt bulb is defective
 - 6 the number of children in a family

Terminology 3: Explanatory and Response Variables

- Often, we want to look at *two* variables *together* and see how their values are *related* or *associated*.
 - ① To study the relationship between two variables, we measure *both* variables on the *same* cases (units, subjects, or individuals).
 - ② For example, if we measure the height and weight of each of a large group of people, we know which height goes with each weight.
- If we are using one variable to help us *understand* or *predict* values of another variable, we call the former the **explanatory (independent, input) variable** and the latter the **response (dependent, output) variable**.
- Notations:
 - ① Explanatory variable = Independent variable = Input variable = X
 - ② Response variable = Dependent variable = Output variable = Y

Exercise 1.2

- A study is conducted to determine if one can predict the yield of a crop based on the amount of yearly rainfall.
 - ① Response variable =
 - ② Explanatory variable =

- A researcher is interested in determining if one could predict the score on a statistics exam from the amount of time spent studying for the exam.
 - ① Response variable =
 - ② Explanatory variable =

Exercise 1.2 (continued)

- It is conjectured that alcohol drops body temperature. To study this, several different amounts of alcohol are given to mice, then the change in each mouse's body temperature is measured in the 15 minutes after taking the alcohol.
 - 1 Response variable =
 - 2 Explanatory variable =
- In study of students' scores on the SAT math exam and the SAT verbal exam, is it more reasonable to simply explore the relationship between the two math and verbal scores or to view one of the scores as an explanatory and the other as a response variable?